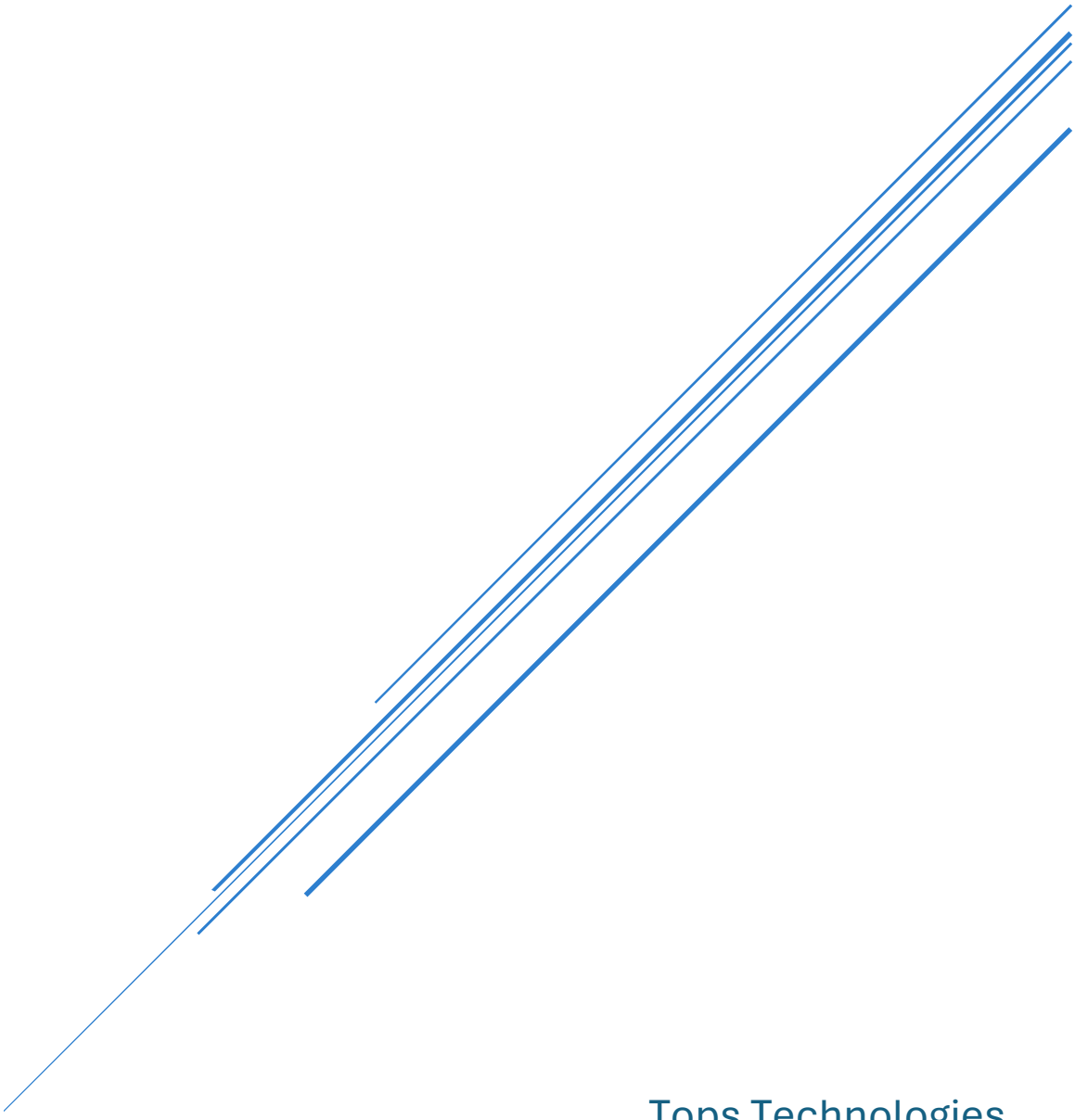
DAY – 2 NETWORK TOPOLOGIES

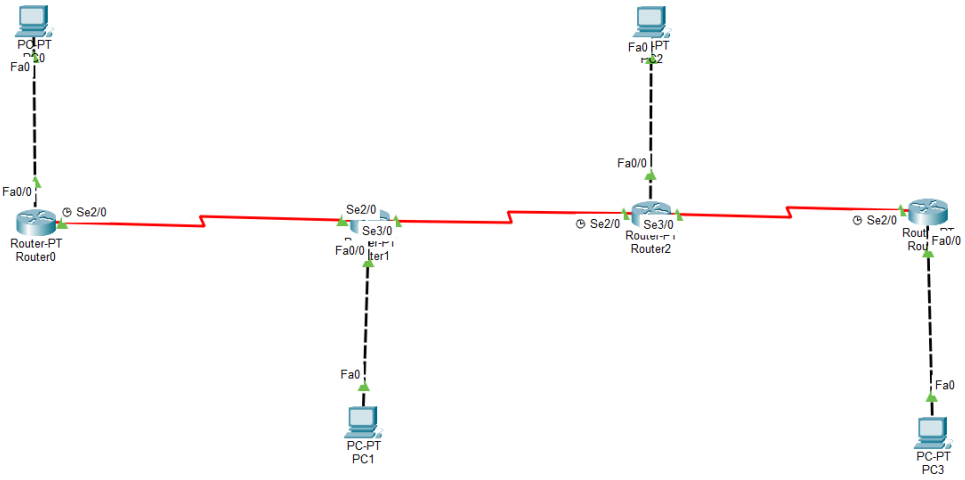
Assignment - 18



Tops Technologies
Cyber Security

1. **Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.**

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1. Description of the Diagram

This network topology is created using Cisco Packet Tracer. It consists of four routers (Router0, Router1, Router2, and Router3) connected in a linear arrangement using serial cables. Each router is connected to one PC through a Fast Ethernet cable. The routers transfer data between different networks, while the PCs send and receive data.

2. Key Components

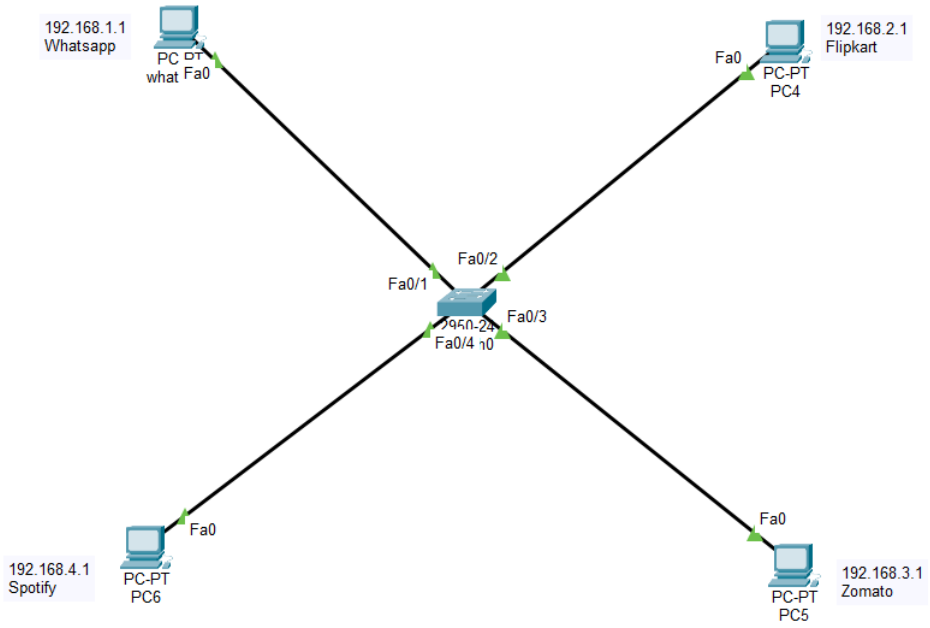
1. **Routers: Four routers connect different networks and forward data.**
2. **PCs: Four PCs act as end devices to send and receive data.**
3. **Serial cables: Red lines connect the routers through serial interfaces.**
4. **Fast Ethernet cables: Black dashed lines connect PCs to routers.**
5. **Interfaces: Fa0/0 represents Fast Ethernet, while Se2/0 and Se3/0 represent serial interfaces.**
6. **Cisco Packet Tracer: Used to design and simulate the network.**

3. Real-World Example

This type of network can be used by a company with branch offices in different cities, such as Ahmedabad, Mumbai, and Delhi. Each branch has a router and computers. The routers communicate through WAN links, allowing employees to exchange data and access shared resources across different branches.

2. **In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.**

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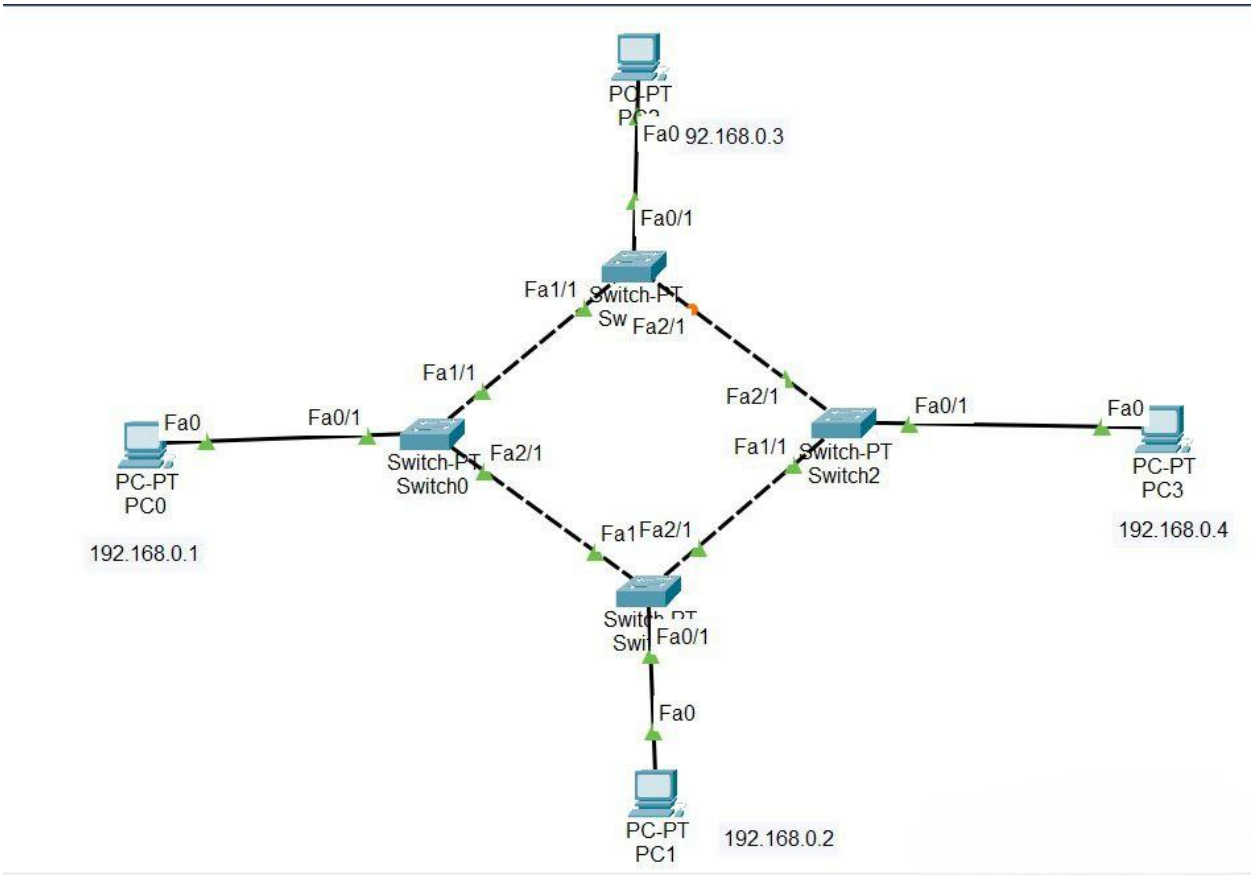


This star topology consists of one central switch connected to four PCs named WhatsApp, Zomato, Spotify, and Flipkart. Each PC is connected directly to the switch using an Ethernet cable. The switch receives data from one device and forwards it to the appropriate destination device. If one cable fails, the other PCs can continue communicating, provided the switch is working properly.

Real-world example: A computer lab where multiple computers are connected to a central switch.

3. **Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.**
   Hint: Use arrows to show the direction of data flow.

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In a ring topology, each device is connected to two neighboring devices, forming a closed loop. Data travels from one device to the next in the direction indicated by the arrows until it reaches the destination device. In a traditional unidirectional ring, each device forwards the data to the next device in the ring.

4. **Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.**

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Topology	Advantages	Disadvantages
Mesh Topology	1. Highly reliable because multiple paths are available for data transmission. 2. Failure of one connection usually does not affect communication through other paths.	1. Requires many cables and network ports, making it expensive. 2. Installation and maintenance are complex.
Star Topology	1. Easy to install, manage, and troubleshoot. 2. Failure of one PC's cable does not affect other connected devices.	1. If the central switch fails, the entire network may stop communicating. 2. Requires more cables than some simpler topologies, increasing installation costs.